

Heisenberg Uncertainty Principle:

It is impossible to know both the exact position and exact momentum of an object (particle) at the same time.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Problem 1: Uncertainty in position of electron $\Delta x = 50 \text{ pm}$, find out the uncertainty in the momentum of the electron?

Solution: Given that uncertainty in the position $\Delta x = 50 \text{ pm}$

$$\Delta x = 50 \times 10^{-12} \text{ m}$$

Now from Heisenberg uncertainty principle

$$\Delta x \Delta p \approx \frac{\hbar}{2}$$

$$\Delta p \approx \frac{\hbar}{2\Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 50 \times 10^{-12}} = 1.05 \times 10^{-24} \text{ kg m/sec}$$

Problem 2: Prove that Heisenberg uncertainty principle is dimensionally correct?

Solution: Since from Heisenberg uncertainty principle we know that

$$\Delta x \Delta p \geq \frac{\hbar}{2} \dots \dots \dots (1)$$

Unit of Δx is meter $\Rightarrow [L]$

Unit of Δp is kg meter/sec $\Rightarrow [MLT^{-1}]$

So dimension of $\Delta x \Delta p = [ML^2T^{-1}]$

$$\text{Dimension of } \hbar = \frac{E}{\omega} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Thus dimensions of equation (1) are balanced and this equation is dimensionally correct.

Problem 3: If the position of an electron in a membrane is measured to an accuracy of $1.00 \mu\text{m}$,

- (a) What is the electron's minimum uncertainty in momentum and velocity?
- (b) If the electron has this velocity, what is its kinetic energy in eV?

Solution: (a) Given that uncertainty in the position $\Delta x = 1 \mu\text{m} = 10^{-6} \text{ m}$

Now from Heisenberg uncertainty principle

$$\Delta x \Delta p \approx \frac{\hbar}{2}$$

$$\Delta p = m \Delta v \approx \frac{\hbar}{2\Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 10^{-6}} = 5.25 \times 10^{-29} \text{ kg m/sec}$$

$$\Delta v \approx \frac{\hbar}{2 m \Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 9.1 \times 10^{-31} \times 10^{-6}} = 57.6 \text{ m/sec}$$

(b) If electron velocity becomes $v = 57.6 \text{ m/sec}$

Then the kinetic energy of the electron

$$KE = \frac{m v^2}{2} = \frac{9.1 \times 10^{-31} \times (57.6)^2}{2} = 1.51 \times 10^{-27} \text{ Joules}$$

$$KE = \frac{1.51 \times 10^{-27}}{1.6 \times 10^{-19}} = 9.43 \times 10^{-9} \text{ eV} = 9.43 \text{ neV}$$

Problem 4: Suppose the velocity of an electron in an atom is known to an accuracy of $2 \times 10^3 \text{ m/sec}$. What is the electron's minimum uncertainty in position?

Solution: Given that uncertainty in the velocity of electron $\Delta v = 2 \times 10^3 \text{ m/sec}$

Therefore the uncertainty in momentum of the electron

$$\Delta p = m \Delta v = 9.1 \times 10^{-31} \times 2 \times 10^3 = 18.2 \times 10^{-28} \text{ kgm/sec}$$

$$\Delta x \approx \frac{\hbar}{2 \Delta p} = \frac{1.05 \times 10^{-34}}{2 \times 18.2 \times 10^{-28}} = 0.0288 \times 10^{-6} = 28.8 \approx 29 \text{ nm}$$

Problem 5: The velocity of a proton in an accelerator is known to an accuracy of 0.250% of the speed of light. What is the smallest possible uncertainty in its position?

Solution: Given that uncertainty in the velocity of electron $\Delta v = 0.25\% \text{ of the velocity of light } c$

$$\Delta v = 0.0025 \times 3 \times 10^8 \text{ m/sec}$$

Uncertainty in momentum

$$\Delta p = m \Delta v = 1.67 \times 10^{-27} \times 0.0025 \times 3 \times 10^8$$

$$\Delta p = 0.012525 \times 10^{-19} \text{ kgm/sec}$$

Uncertainty in position

$$\Delta x \approx \frac{\hbar}{2 \Delta p} = \frac{1.05 \times 10^{-34}}{2 \times 0.012525 \times 10^{-19}} = 4.19 \times 10^{-15}$$

Problem 6: Show that the particle which has uncertainty in position $\Delta x = \frac{\lambda}{2\pi}$ where λ is de-Broglie wavelength. Show that velocity uncertainty is same order of magnitude as the velocity itself.

Solution: Since from Heisenberg uncertainty principle we can write

$$\Delta x \Delta p \approx \frac{\hbar}{2} \dots \dots \dots (1)$$

Given that uncertainty in position $\Delta x = \frac{\lambda}{2\pi}$

Thus the uncertainty in velocity $\Delta v \approx \frac{\hbar}{2 m \Delta x} = \frac{\hbar \times 2\pi}{2 m \lambda} = \frac{h}{2m\lambda} \dots \dots \dots (1)$

From de-Broglie wavelength relation

$$\lambda = \frac{h}{mv}$$

$$v = \frac{h}{m \lambda} \dots \dots \dots (2)$$

Thus from equation (1) and (2) the velocity and uncertainty in velocity, are same order of magnitude.